

The Equation-of-Motion Hinge

From physical law to differential equation to one particular motion

IB Physics A3 worksheet

Central idea. A physical law gives a rule for the instantaneous change of a system. Written with derivatives, it becomes a differential equation. A proposed function of time is a solution when its derivatives satisfy that equation. Initial conditions select the particular motion.

physical model $\longrightarrow$ differential equation $\longrightarrow$ family of solutions $\longrightarrow$ initial conditions .

For each panel, distinguish the **equation of motion** from its **solution**. Verify the solution by differentiating; formal integration is not required.

1. Constant gravitational field

Take upward as positive. Near Earth’s surface,

$$F_y = -mg.$$

Newton’s second law gives

$$m \frac{d^2y}{dt^2} = -mg \quad \Rightarrow \quad \frac{d^2y}{dt^2} = -g.$$

A proposed solution is

$$y(t) = y_0 + u_y t - \frac{1}{2}gt^2.$$

Tasks

1. Differentiate twice and verify that the proposed solution satisfies the equation of motion.

2. Identify the two initial conditions contained in the solution.

3. For $y_0 = 1.5 \text{ m}$ and $u_y = 8.0 \text{ m s}^{-1}$, write the particular solution.

4. Explain why $d^2y/dt^2 = -g$ does not by itself specify one trajectory.

Hinge question Which information comes from the gravitational model, and which information comes from the launch?

2. Simple harmonic motion

For an ideal mass–spring system,

$$F = -kx.$$

Newton’s second law gives

$$m \frac{d^2x}{dt^2} = -kx,$$

so

$$\frac{d^2x}{dt^2} + \omega^2x = 0, \quad \omega = \sqrt{\frac{k}{m}}.$$

A proposed solution is

$$x(t) = A \cos(\omega t + \phi).$$

Tasks

1. Differentiate twice and verify that the proposed solution satisfies the equation of motion.

2. State what determines ω .

3. State what determines A and ϕ .

4. A system has $x(0) = A$ and $v(0) = 0$. State a suitable value of ϕ .

5. Explain why the equation predicts oscillation rather than exponential decay.

Hinge question What is fixed by the apparatus, and what is fixed by how the system is released?

3. Exponential change

Radioactive decay is modelled by

$$\frac{dN}{dt} = -\lambda N.$$

A proposed solution is

$$N(t) = N_0 e^{-\lambda t}.$$

Capacitor discharge has the parallel form

$$\frac{dQ}{dt} = -\frac{Q}{RC}, \quad Q(t) = Q_0 e^{-t/RC}.$$

Tasks

1. Differentiate $N(t)$ and verify that it satisfies the decay equation.

2. Explain the physical meaning of the minus sign.

3. Identify the initial condition contained in $N(t)$.

4. Match the constants in the two equations:

$$\lambda \quad \longleftrightarrow \quad \text{_____}.$$

5. State why these equations are first order while the motion equations in Panels 1 and 2 are second order.

Hinge question What shared mathematical rule produces an exponential solution in both systems?

Synthesis. Complete the architecture for each system.

System	Equation of motion/change	Form of solution	What selects the particular solution?
Constant gravitational field			
Mass–spring oscillator			
Radioactive decay or capacitor discharge			

Complete: An equation of motion specifies _____. A solution specifies _____. Initial conditions specify _____.

The Equation-of-Motion Hinge

Teacher guide

Purpose. Students separate four levels that are often compressed into one formula: the interaction model, its differential equation, a family of functions satisfying that equation, and the initial conditions selecting one member of that family. Verification by differentiation keeps the treatment within IB mathematical expectations.

1. Constant gravitational field

Starting from

$$y(t) = y_0 + u_y t - \frac{1}{2}gt^2,$$

we obtain

$$\frac{dy}{dt} = u_y - gt, \quad \frac{d^2y}{dt^2} = -g.$$

The initial conditions are

$$y(0) = y_0, \quad v_y(0) = u_y.$$

For the given data,

$$y(t) = 1.5 + 8.0t - 4.905t^2.$$

The differential equation fixes the curvature of every allowed trajectory. Different values of y_0 and u_y produce different trajectories satisfying the same equation.

Model versus launch:

- the field model supplies g and the equation $y'' = -g$;
- the launch supplies initial position and velocity.

2. Simple harmonic motion

For

$$\begin{aligned} x(t) &= A \cos(\omega t + \phi), \\ \frac{dx}{dt} &= -A\omega \sin(\omega t + \phi), \\ \frac{d^2x}{dt^2} &= -A\omega^2 \cos(\omega t + \phi) = -\omega^2 x. \end{aligned}$$

Therefore

$$x'' + \omega^2 x = 0.$$

The apparatus parameters determine

$$\omega = \sqrt{\frac{k}{m}}.$$

The initial displacement and velocity determine A and ϕ . For $x(0) = A$ and $v(0) = 0$, one suitable choice is $\phi = 0$.

Oscillation follows from acceleration being opposite and proportional to displacement:

$$a = -\omega^2 x.$$

The system is repeatedly accelerated back toward equilibrium.

3. Exponential change

For

$$N(t) = N_0 e^{-\lambda t},$$

$$\frac{dN}{dt} = -\lambda N_0 e^{-\lambda t} = -\lambda N.$$

The minus sign states that the quantity decreases. The initial condition is

$$N(0) = N_0.$$

For capacitor discharge,

$$\lambda \longleftrightarrow \frac{1}{RC}.$$

Both equations state that the instantaneous rate of decrease is proportional to the amount remaining.

The decay and discharge equations are first order because their highest derivative is a first derivative. Mechanical motion under Newton’s second law is normally second order because acceleration is the second derivative of position. Consequently, one initial condition selects a first-order solution, whereas position and velocity are needed for a second-order motion equation.

Synthesis answers

System	Equation	Solution form	Selection data
Uniform gravity	$y'' = -g$	$y = y_0 + u_y t - \frac{1}{2}gt^2$	y_0, u_y
Mass–spring	$x'' + \omega^2 x = 0$	$x = A \cos(\omega t + \phi)$	initial x and v
Decay/discharge	$N' = -\lambda N$, or $Q' = -Q/RC$	exponential	initial amount N_0 or Q_0

Desired final statements

- An equation of motion specifies how the state must change at each instant.
- A solution specifies a complete possible history through time.
- Initial conditions select the history followed by the particular system.

Misconception to target. A formula such as $y = y_0 + u_y t - \frac{1}{2}gt^2$ is already a solved trajectory. The underlying equation of motion is $y'' = -g$. Conversely, writing $a = -g$ does not yet identify the object’s position or velocity at any time.